

Detecting depression from *interview speech.*

*Classifying PHQ-8 severity from DAIC-WOZ
transcripts.*

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BACKGROUND

THE PROBLEM

Depression is *underdiagnosed* — and the symptoms often surface in how people *talk*.

Clinicians screen with the PHQ-8, an 8-item questionnaire scored 0–24.

If text alone can flag risk, screening can scale beyond the clinic.

Prior NLP work has shown signal in word choice, sentiment, and disfluencies.

THE PROJECT GOAL

Can we predict a participant's PHQ-8 severity range from the words in their interview?

Each member binarizes the same dataset on a different cutoff, then compares classical machine learning with a transformer-based model.

Each member owns one PHQ-8 band.

GRANT · 0–4

None /
minimal

Class 1 = score in 0–4. The "healthy" cutoff.



0 · 77 (54.5%) 1 · 64 (45.4%)

ERIK · 5–9

Mild

Class 1 = score in 5–9. Subclinical territory.



0 · 106 (75.2%) 1 · 35 (24.8%)

JACKSON · 10–14

Mild–
moderate

Class 1 = score in 10–14. Clinical threshold begins.



0 · 117 (83.0%) 1 · 24 (17.0%)

SIMON · 15+

Moderate–
severe

Class 1 = score ≥ 15. Highest risk band.



0 · 123 (87.2%) 1 · 18 (12.8%)

DATASET

DAIC-WOZ clinical interviews.

SOURCE

USC ICT Distress Analysis
Interview Corpus

Semi-structured interviews conducted
by an animated virtual interviewer
("Ellie"). Each participant has a
transcript and a PHQ-8 score.

PARTICIPANTS

141

After cleaning

MODALITY

Transcript only

Text features for this project

Classical sweep, then a transformer.

TRACK A

8 classical models

- TF-IDF + PCA features
- minority class upsampled
- 8 classifiers (RF, NB, kNN, LR, 2× SVM, AdaBoost, XGBoost)
- 15 random splits
- evaluated on balanced accuracy, sensitivity, specificity.

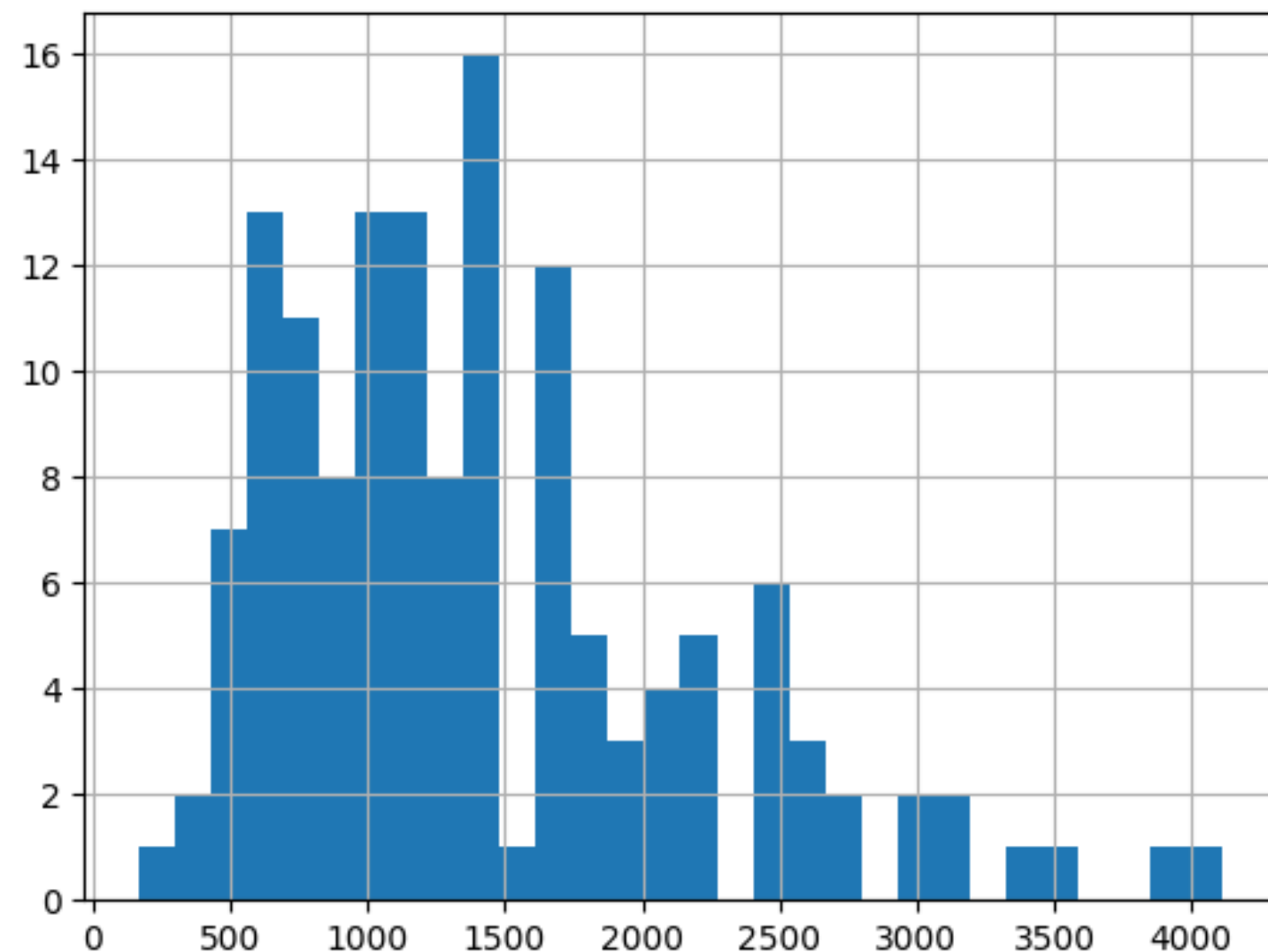
TRACK B

BERT fine-tune

- pretrained BERT (bert-base-uncased)
- 512-token transcript limit
- frozen encoder + 2-layer MLP head
- class-weighted NLL loss
- learning-rate sweep (1e-7, 5e-5)

TOKEN LENGTHS

Most transcripts exceed BERT's 512-token max length.



MEDIAN TOKENS

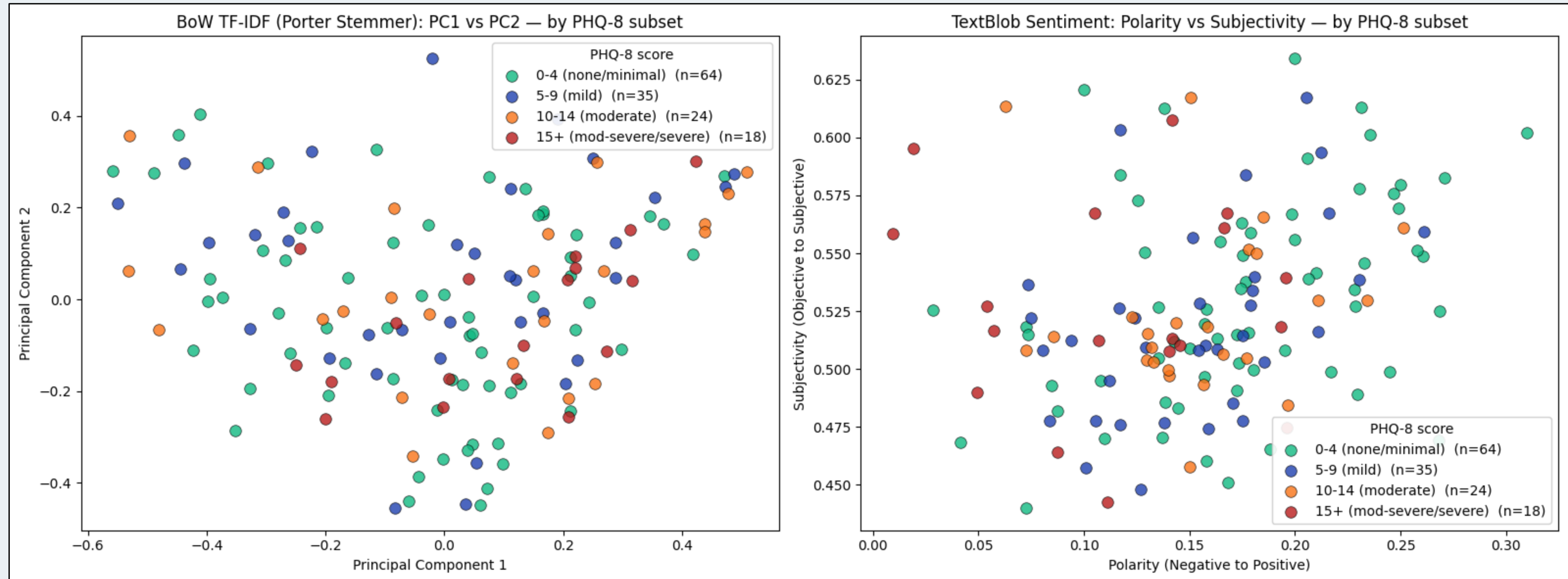
~1,300

Per cleaned transcript

Distribution is right-skewed — most interviews land between 500 and 2,000 tokens, with a long tail to ~4,000.

For BERT we set the max token length to 512 tokens, which captures the densest part of the distribution.

Severity does not separate cleanly in 2D.



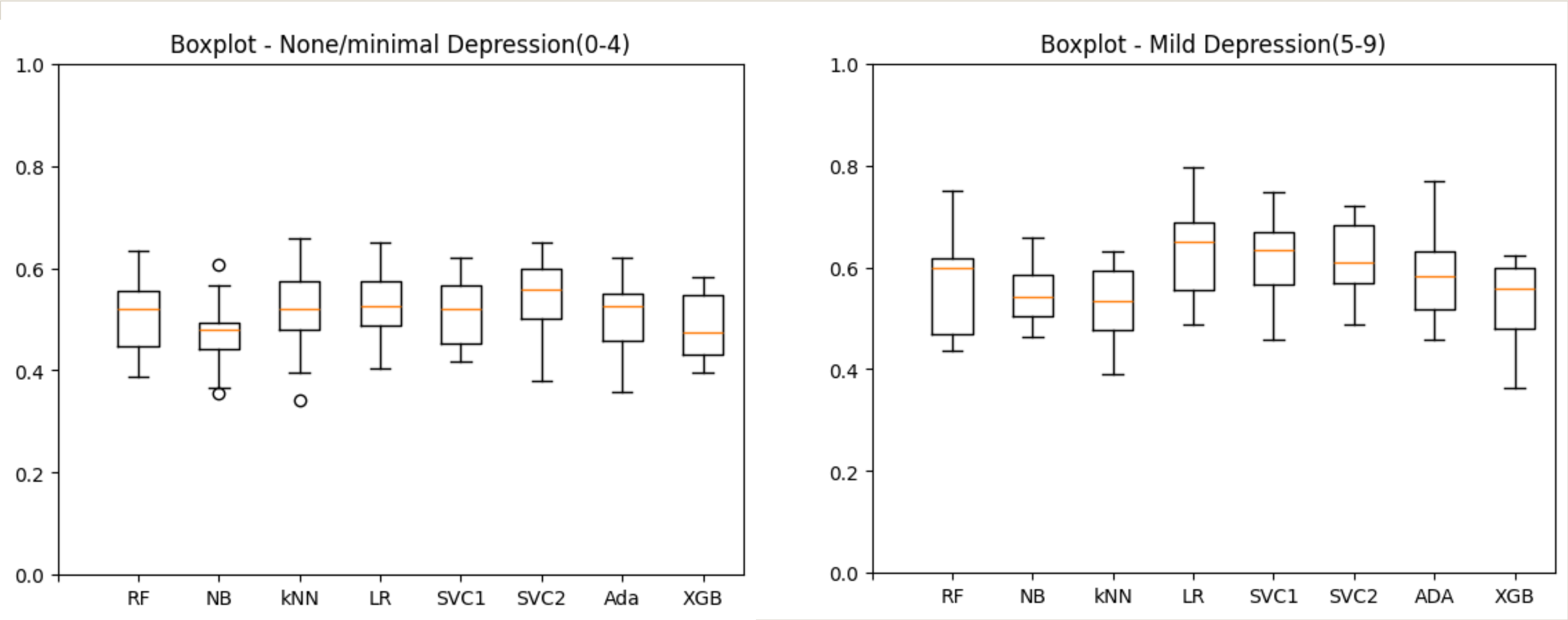
PC1 × PC2 WITH ALL BANDS HIGHLIGHTED

Each subset is distinctly labeled.

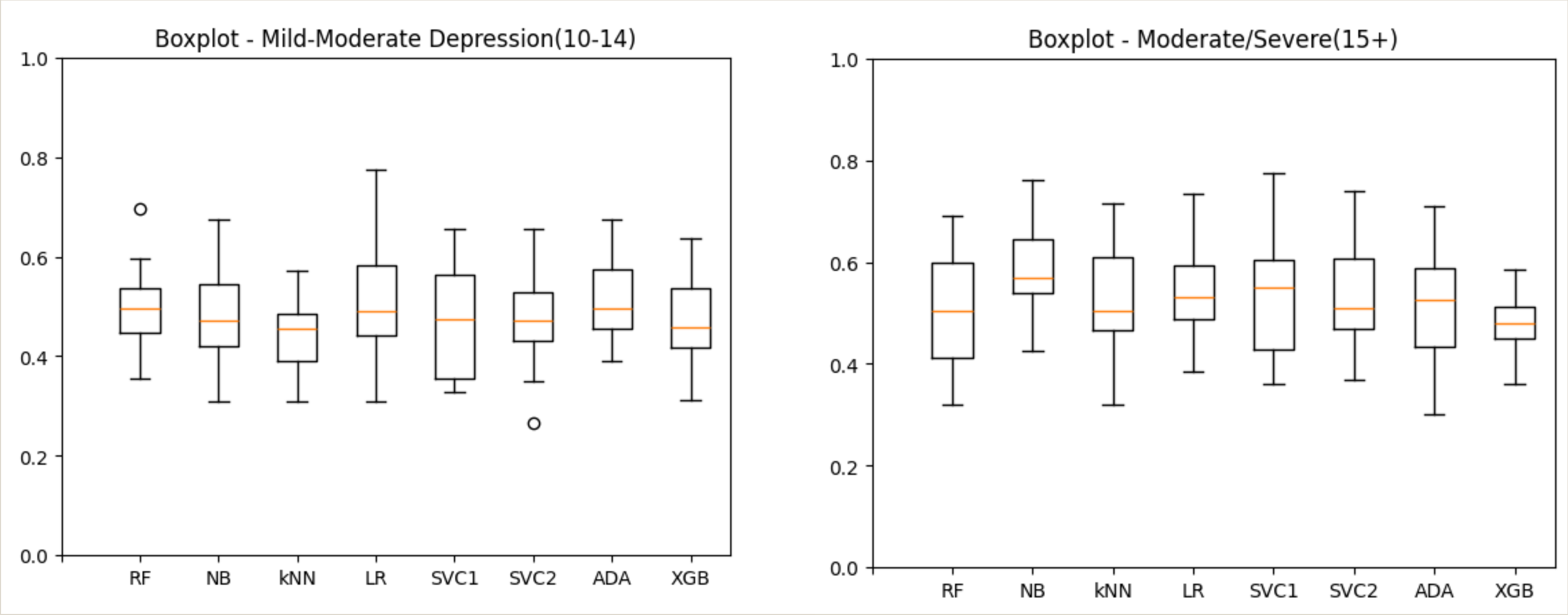
The subsets are scattered through the cloud, not clustered. Linear decision boundaries struggle by design here.

2 Principal Components capture only 21% of the variance

Most models cluster around 0.50 — chance.



Most models cluster around 0.50 — chance.



No / minimal symptoms — the most balanced split.

BEST MODEL

LR / SVC2

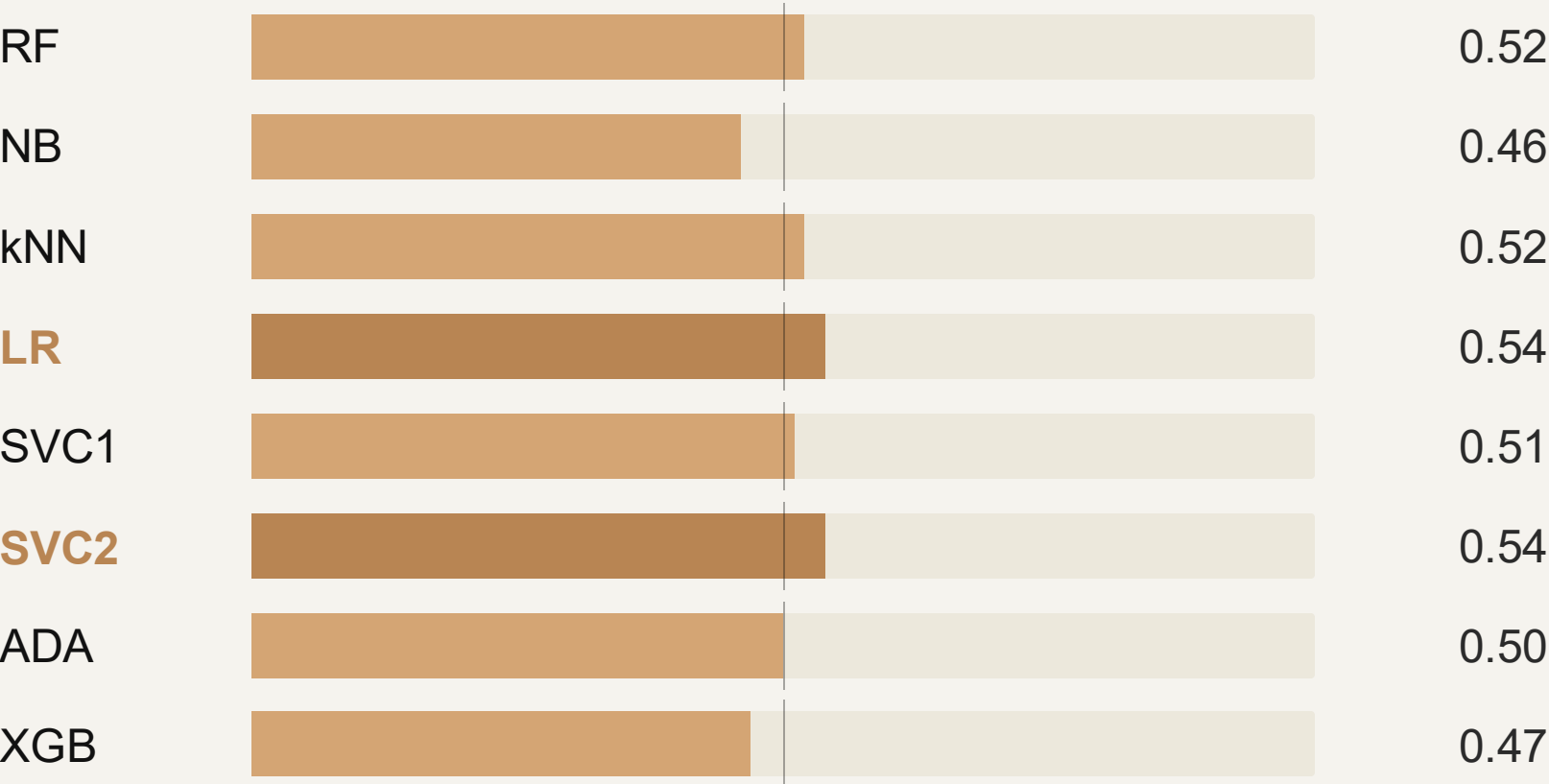
LR sens 0.54 · spec 0.53 · tied with linear SVM

BALANCED ACCURACY

0.54

±0.07 — modest lift on a near-balanced task

Even with the most balanced classes (45% positive), shallow features only push 4 points above chance.



Mild symptoms — most learnable band.

BEST MODEL

LR

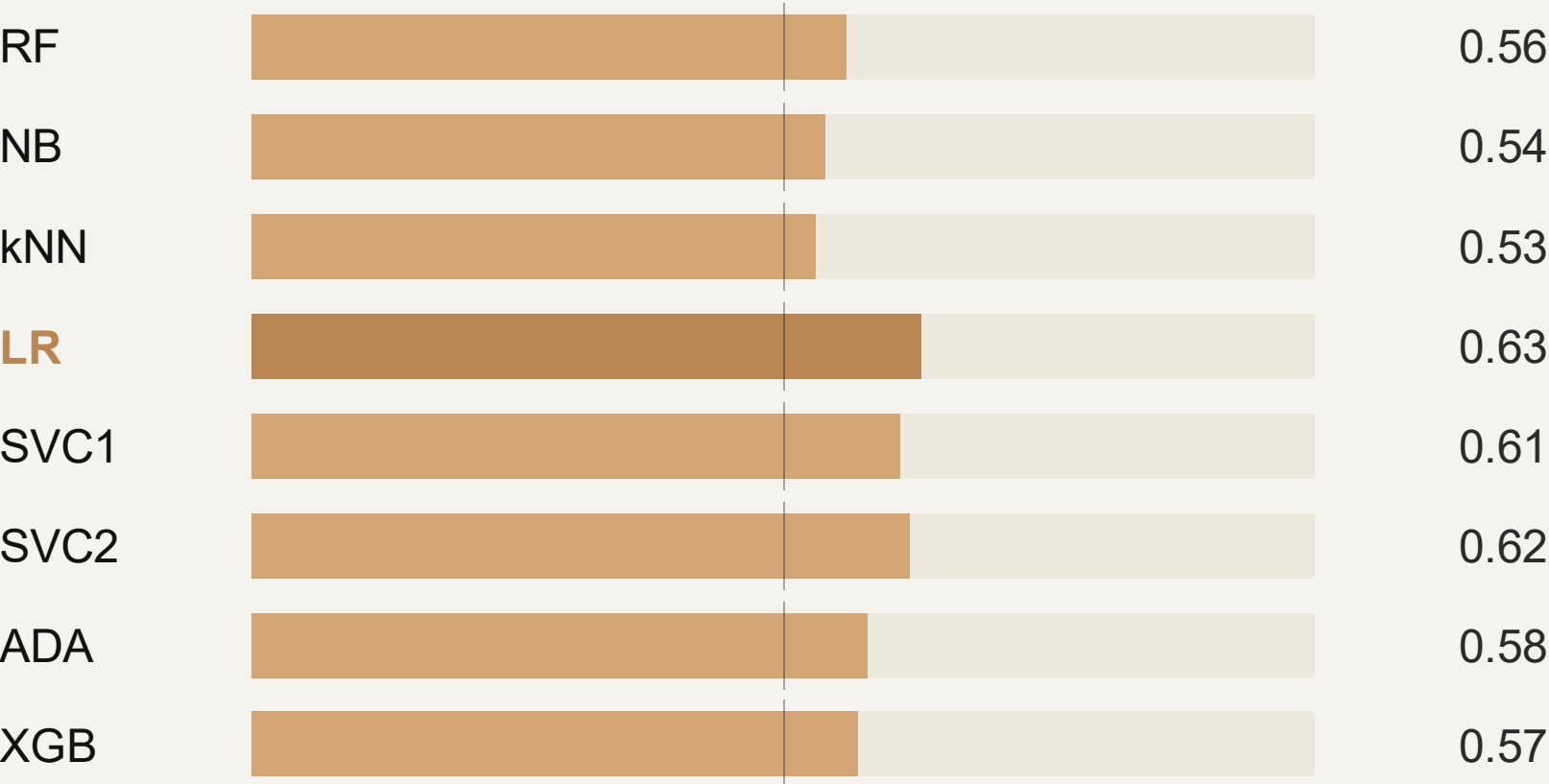
Sensitivity 0.59 · Specificity 0.66

BALANCED ACCURACY

0.63

±0.09 across 15 random splits

Regularized linear models lead. Classes are closer to balanced here, which helps every model.



Mild–moderate symptoms — a noisy middle band.

BEST MODEL

ADA

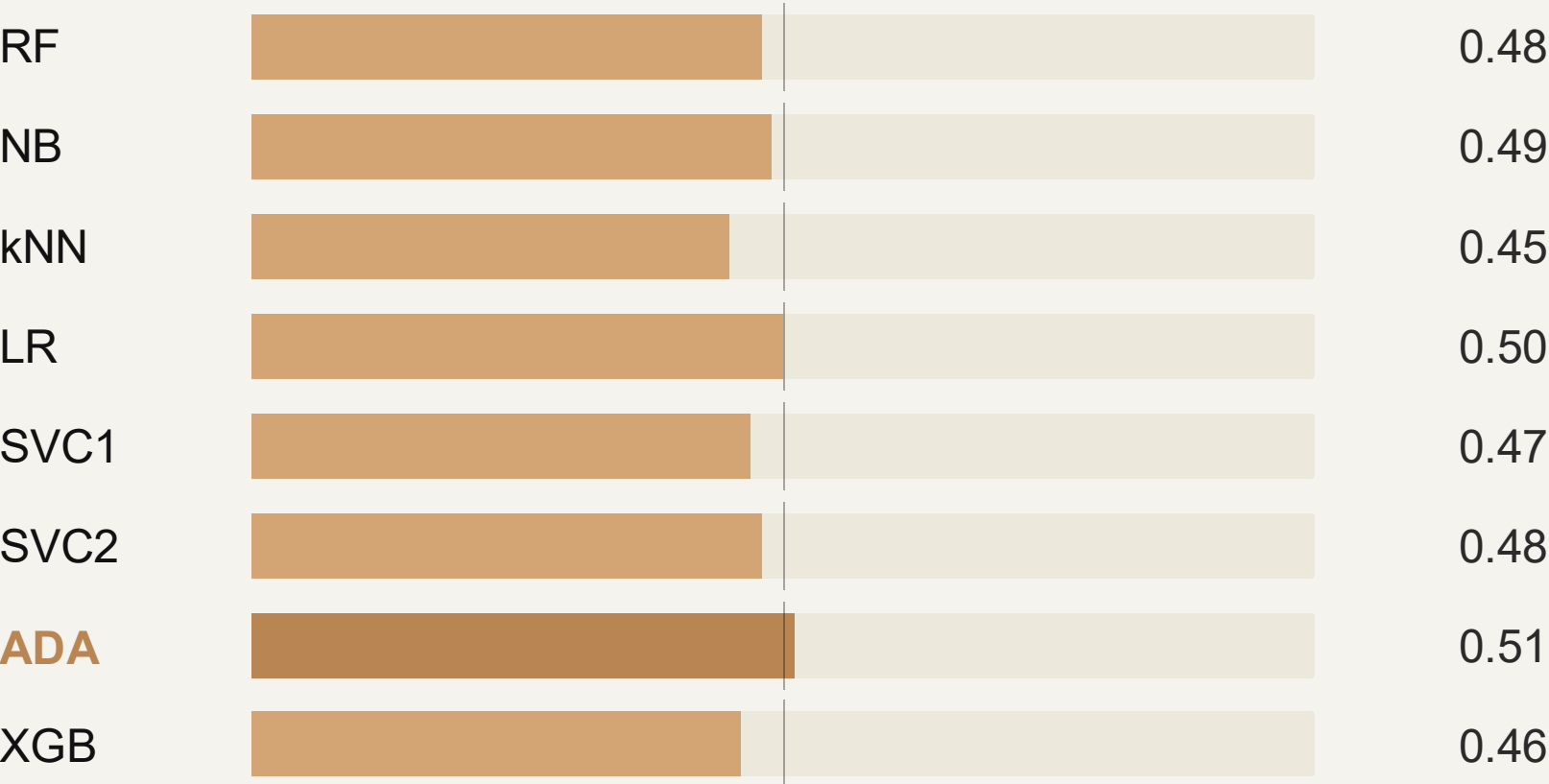
Sensitivity 0.31 · Specificity 0.71

BALANCED ACCURACY

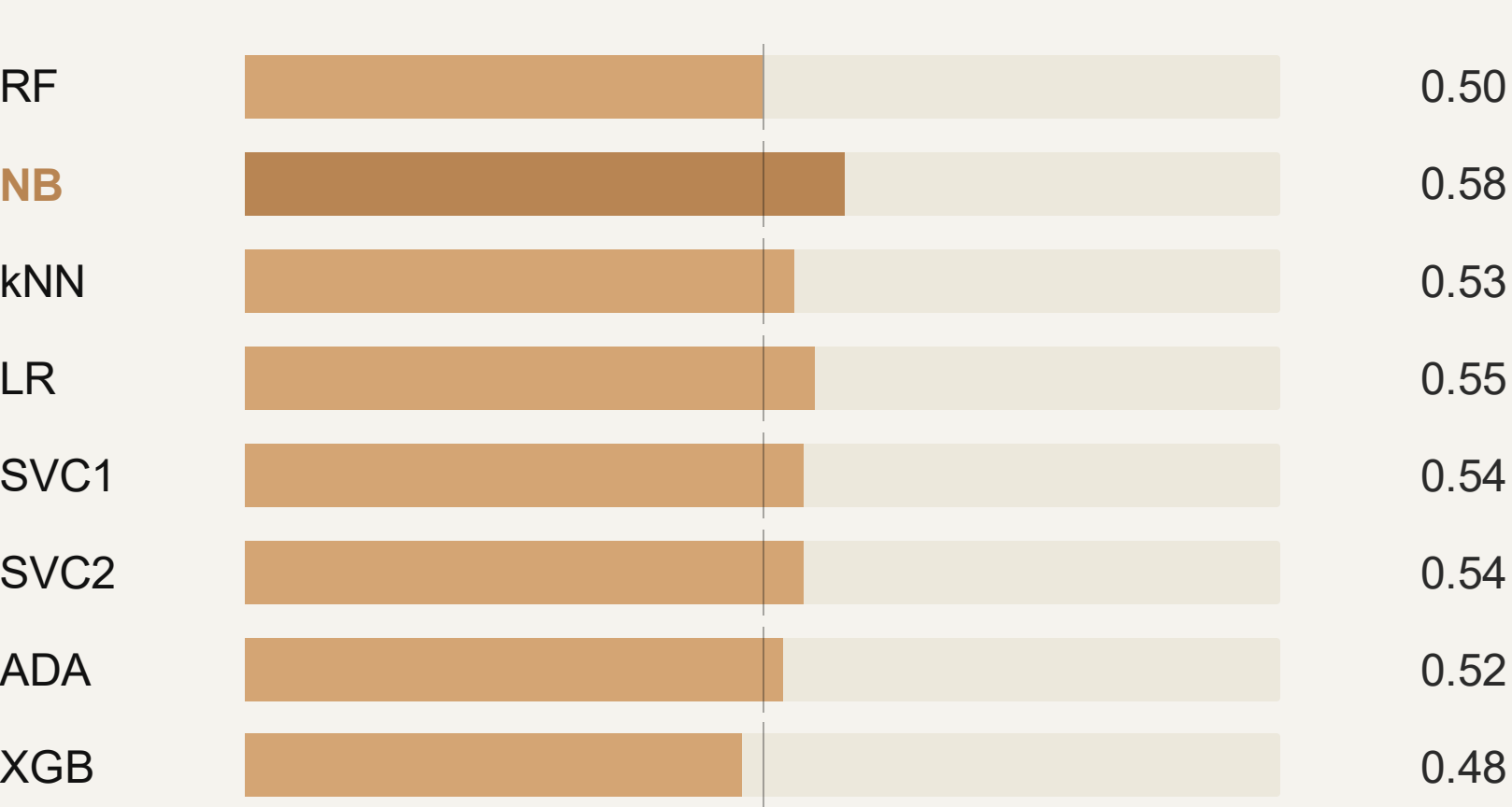
0.51

±0.07 — barely above chance

17% positive rate. Mild–moderate participants speak too similarly to the rest for shallow features to separate them.



Moderate–severe symptoms — Naive Bayes wins.



BEST MODEL

NB

Sensitivity 0.6 · Specificity 0.563

BALANCED ACCURACY

0.58

±0.09 · only 12.8% positive class

Different bands, different winners.

MEMBER	PHQ-8 BAND	CLASS 1 SHARE	BEST MODEL	BAL. ACC.	SENS.	SPEC.
Grant	0–4 (none)	45.4%	LR / SVC2	0.54	0.54	0.53
Erik	5–9 (mild)	24.8%	LR	0.63	0.59	0.66
Jackson	10–14 (mild–mod.)	17.0%	ADA	0.51	0.31	0.71
Simon	15+ (mod.–severe)	12.8%	NB	0.56	0.60	0.56

Erik's mild band is the clear winner — likely because its 75/25 split is hard but not impossible for shallow text features. The extreme bands (Jackson, Simon) hover near chance.

BERT didn't beat the simple baseline.

BEST CLASSICAL (ERIK · LR)

0.63

Logistic regression on TF-IDF + PCA, 15 runs.

BEST BERT (JACKSON · LR SWEEP)

0.50

Across $LR \in \{1e-7, 1e-6, 5e-6, 1e-5, 5e-5\}$, three trials collapsed to majority-class prediction (ba = 0.50, sens = 0.0, spec = 1.0).

With ~98 training examples and ~22 positive cases, the frozen-encoder + small head couldn't escape the class imbalance. The model that "looked best" by accuracy simply predicted "not depressed" every time.

Limitations

01

Small dataset

Only 141 participants after cleaning
Deep models like BERT struggled
with limited training data.

02

Class imbalance

Some PHQ-8 bands had very few
positive examples
Models often defaulted to majority-
class predictions.

03

Text only

Audio and behavioral cues were
excluded
Important signals like pauses, tone,
and hesitation were lost.

04

Limited PCA variance

Two principal components captured only a small share of total feature variance (21%)
This limited the models' ability to explain the variance present in the TF-IDF feature space

Four directions worth pursuing.

- 01 Train BERT on the **full PHQ-8 regression** rather than a per-member binary cutoff — more signal per example.
- 02 Unfreeze the encoder and add early stopping; the frozen + head setup couldn't adapt to clinical language.
- 03 Add audio + acoustic features from the original DAIC recordings — pauses and prosody likely carry the signal disfluencies hint at.
- 04 k-fold cross-validation in place of repeated random splits — tighter variance estimates on a small dataset.

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Thank you.

QUESTIONS

Grant · Erik · Jackson · Simon